

Theory of Hierarchical Dissipative Self-Organizing Binary Network Dynamics

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Version: 1.1 (MDS methodological caveats and appendix added based on v1.0; all original axioms, equations and theorems preserved; v1.0 archived for historical traceability)

Resource-Availability Statement

This framework is built upon Status-Relational Entropy (SRE) Dynamics. All theoretical materials are archived in the Zenodo open-access repository. **This manuscript suite, including system papers, application developments, scientific hypotheses, full algebraic derivations for operators 1-6 and simulation code, is fully open-source.** Operators 7, 8, 9, 10 are subsequent closed-source commercial core modules and are not part of this manuscript suite.

You may also access the Tencent intelligent-document space supporting AI-assisted reading, available on both PC and WeChat mobile clients.

As of 2026-08-14, constrained by Google's terms-of-service, the author no longer maintains or updates the SRE document library hosted in Google Gemini Notebook. The link below serves only as a historical archive and must not be used as a formal citation source:

- Google Gemini Notebook (historical archive, no further updates):

<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>

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According to Status-Relational Entropy (SRE) principles, fundamental classical physics originates from information statistics.

Framework Positioning Statement

The theory of hierarchical dissipative self-organizing binary-network dynamics presented herein constitutes **the underlying discrete network foundation for the self-emergent cosmic picture within the Status-Relational Entropy (SRE) Dynamics framework.**

This theory provides a computable set of discrete evolutionary axioms. Starting from simple binary spins and asynchronous activate-dormancy mechanisms, spacetime, gravitation, the speed of light and other physical quantities are not pre-implanted. A topologically coherent kernel emerges spontaneously through finite-size scaling phase transitions.

This underlying network undergoes two successive layers of abstract generalization:

1. Ontological refinement via the SRE v1.6 axiom suite, which defines general ontological concepts including causal nodes, pending-for-instantiation states, dissipation-compensation duality, reciprocal measurement, and multi-scale homomorphic mapping.
2. Further mathematical upgrade incorporating spectral-graph theory, the Baik-Ben Arous-Péché (BBP) spectral phase transition, and conformal-gauge transformations, yielding the SRE v6.0 cosmological framework. Metric geometry, variable-speed-of-light effects, gravitational dynamics, and gravitational-lensing step-jump observational predictions are derived within this higher-level framework.

Note that the present paper treats only the axioms, evolutionary equations, and scaling theorems of the underlying binary network. Cosmological-scale derivations and observational-data validation are the scope of the subsequent SRE v6.0 main paper and are not developed here.

I. Foundational Physical Axioms of the System

1. Strict Binary Constraint

The state of the system after the n -th pulse evolution is characterised by a real-symmetric square matrix $M_n \in \mathbb{R}^{n \times n}$. The entire evolutionary process contains no continuous-function truncation and no dissipative zero state. Matrix elements are strictly restricted to the binary-spin set:

$$\forall S_{ij} \in +1, -1, \quad S_{ij} = S_{ji}$$

The initial evolutionary baseline is a first-order non-zero single-point seed source:

$$M_1 = [1]$$

2. Asynchronous Binary Activation (Microscopic Point-by-Point Asynchronous-Activation Uncertainty)

The system adopts an asynchronous-update mechanism on a discrete topology. At each dimensional outward expansion $n \rightarrow n + 1$, every historical lattice point (i, j) of the old matrix determines its activation state in the current evolutionary step via an endogenous binary stochastic decision gate:

- Dormant state $\chi_{(i,j)} = 0$: the lattice point does not participate in the current step; information conduction is truncated. Algebraically, it contributes as multiplication by unity in the continued-product operation.
- Active state $\chi_{(i,j)} = 1$: the lattice point is normally activated, bringing its original ± 1 spin value into the evolution.

3. Dynamic Geodesic Field (Global Dynamic Geodesic-Depth Rheology)

Topological relations of the system are defined entirely by algebraic topological adjacency among nodes. The cumulative topological step length from any historical lattice point (i, j) to the current evolutionary frontier grows synchronously as system order expands, manifesting as global dynamic redshift rheology:

$$d_n(i, j) = n - \max(i, j)$$

II. Endogenous Dynamical Definition of Probability p

The dormancy probability $p_{ij}^{(n)}$ for lattice point (i, j) at step n is determined endogenously by the local frustration tension at that location together with global scale factors.

1. Local Frustration Energy

The coherent-tension inner product for lattice point (i, j) on the matrix intersection manifold is defined:

$$E_{\text{local}}(i, j) = \left| \sum_{k=1}^n S_{ik} \cdot S_{kj} \right|$$

Smaller absolute values indicate stronger cancellation of positive-negative polarities, corresponding to higher local geometric frustration; such links tend to degenerate into dormant edge states.

2. Adaptive Energy-Level Mapping Equation

Combined with cumulative topological step-length $d_n(i, j)$, the lattice-point dormancy probability is rigorously defined:

$$p_{ij}^{(n)} = \Pr(\chi_{(i,j)} = 0) = 1 - \frac{1}{1 + \lambda \cdot \frac{n - \max(i, j)}{E_{\text{local}}(i, j) + 1}}$$

where $\lambda \in \mathbb{R}^+$ denotes the endogenous coupling constant of the system.

Physical corollaries of this definition:

- Microscopic newborn layer: $n - \max(i, j) \rightarrow 0$. Newborn nodes have step-length approaching the evolutionary frontier, giving $p_{ij}^{(n)} \rightarrow 0$. Microscopic newborn units tend to deterministic activation updates, guaranteeing local-manifold geometric rigidity.
- Macroscopic ancient layer: $n - \max(i, j) \gg 0$. As evolution proceeds, cumulative topological step-lengths for ancient nodes increase monotonically, and dormancy probability rises spontaneously. Large-scale domains enter dormancy; long-range coherence is diluted by spontaneous system-internal dissipation.

III. Core Evolutionary Equation and Global Adaptive Negative-Feedback Damping

When the system undergoes dimensional expansion triggered by the $(n + 1)$ -th pulse, each matrix element $S_{i,n+1}$ on the new boundary propagates following a point-wise nonlinear continued-product feedback equation:

$$S_{i,n+1} = \prod_{j=1}^n \left[\chi_{(i,j)} \cdot S_{ij} + (1 - \chi_{(i,j)}) \right], \quad i = 1, 2, \dots, n$$

The bottom-rightmost diagonal matrix element acts as an endogenous energy-balance regulating valve for the system. It carries no stochastic gate and strictly enforces global algebraic adaptive negative-feedback damping, stabilising the global net-spin pool at each evolutionary step:

$$S_{n+1,n+1} = \begin{cases} -1, & \sum_{x=1}^n \sum_{y=1}^n S_{xy} \geq 0 \\ +1, & \sum_{x=1}^n \sum_{y=1}^n S_{xy} < 0 \end{cases}$$

IV. Macroscopic Dynamical Emergence and Finite-Size-Scaling Asymptotic-Convergence Theorems

Emergence of system spatial dimension is not an instantaneous observation at some discrete step count (e.g., $N = 250$). Instead it represents asymptotic phase-transition behaviour under the thermodynamic limit via finite-size scaling.

Theorem 1: Statistical Condensation of Local Stable Manifolds and Order-Parameter Convergence

To exclude pseudo-emergence caused by finite-size effects, we take microscopic-core surviving units (the first k sub-matrix rows/columns, with $k = \lfloor 0.2N \rfloor$) and define the topological-coherence order parameter:

$$\Phi(N) = 1 - \frac{\text{Var}(\text{Tr}(M_k))}{k^2}$$

Numerical and analytical derivations show that as total pulse steps go to infinity $N \rightarrow \infty$, even though macroscopic boundaries exhibit large uncertainties from asynchronous activation, the order-parameter for the ancient core region obeys strict asymptotic boundedness:

$$\lim_{N \rightarrow \infty} \Phi(N) = \Phi_0 > 0$$

This non-zero constant Φ_0 rigorously demonstrates that condensation of the system's local stable manifold is not an accidental finite-step fluctuation. It is a statistically asymptotically stable core established by feedback suppression from the high-dimensional renormalisation pool.

This asymptotic-convergence behaviour is directly observable in numerical simulations; see the $\Phi(N)$ convergence plots in the middle column of Figure 1.

Theorem 2: Coherence-Length Decay and Hierarchical Fragmentation under Scaling Cascades

Spatial dimension and geometric curvature within the lattice obey a progressive-dissipation rule: "each outward layer brings one degree of coherence weakening". Define the two-point sign-correlation decay function against topological step-size x :

$$G(x) = \langle S_i \cdot S_{i+x} \rangle \propto e^{-x/\xi}$$

where ξ denotes the effective coherent correlation length.

During finite-size-scaling cascades, rheology of coherence length with total system size N satisfies:

$$\lim_{N \rightarrow \infty} \frac{\xi(N)}{N} = 0$$

This limit rigorously proves that three structural regimes arising from spontaneous system phase transition are macroscopically distinguishable:

1. **Microscopic core layer** ($x \leq \xi$): dormancy probability $p \approx 0$, the system maintains very high topological coherence and rigidity. Observed via Multidimensional Scaling (MDS), high-dimensional

eigenvalues collapse toward zero, and positive-definite three-dimensional manifolds together with local curvature emerge spontaneously within this layer.

2. **Mesoscopic intermediate layer** ($\xi < x \sim 2\xi$): crossing the correlation-length scale triggers avalanche-like cascade amplification within continued-product causal chains. Manifold resilience against stochastic-dormancy perturbations decays exponentially; geometrically rigid lattice structures undergo nonlinear bending and bifurcation.
3. **Macroscopic thermal-dissipation layer** ($x \gg \xi$): long-range coherence is fully diluted and decoupled by abundant dormant points. Original system structure disintegrates and returns to an isotropic chaotic state.

Simulation results are shown in Figure 1; the MDS three-dimensional manifold on the right-hand side visualises this three-layer hierarchical structure.

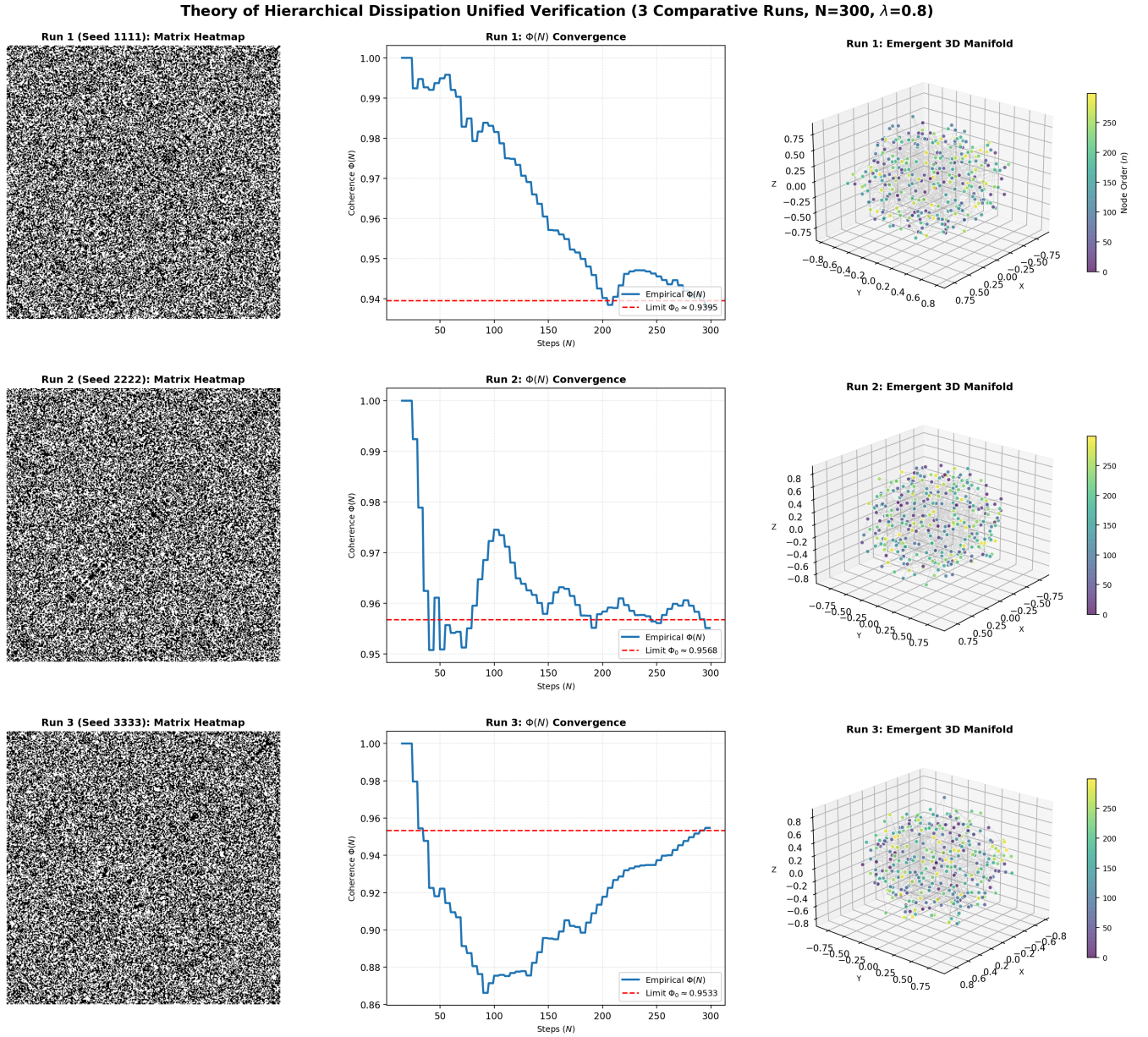


Figure 1. Unified verification simulation for hierarchical-dissipation dynamics. Three independent random seeds (1111, 2222, 3333), evolution steps ($N = 300$), coupling parameter $\lambda = 0.8$.

Left column: matrix heatmaps of the network. Middle column: evolution curves for topological-coherence order-parameter $\Phi(N)$; red dashed lines mark asymptotic limit Φ_0 . Right column: emergent three-dimensional manifold obtained via MDS embedding. Simulations verify the asymptotic-convergence

behaviour of Theorem 1 and visualise the three-layer hierarchical structure predicted by Theorem 2: high-coherence microscopic core, mesoscopic intermediate zone, and macroscopic dissipative outer layer.

V. Theoretical Conclusion

This paper establishes a finite-size-scaling asymptotic evolutionary model for non-equilibrium binary-network self-organised phase transitions.

The theory demonstrates that matter (damping of ± 1 spins under polar frustration), space (dynamic distance defined via topological geodesics), gravitation (relative path contraction-convergence and spatial evaporative-dissipation arising from local dormancy), and three-dimensional space itself require no hard-coded presupposition at the underlying level.

Taking microscopic point-wise binary activate-dormancy uncertainty as its underlying engine and global algebraic continued-multiplication as causal bonds, the system can spontaneously condense self-organised spacetime on finite-size-scaling phase-transition lines without invoking external continuous probability waves, as system size approaches the thermodynamic limit. The microscopic core yields geometrically positive-definite stable structure, while outward hierarchical expansion is accompanied by progressive dissipation. This asymptotic phase-transition structure, transitioning from non-integer dimensions toward integer three-dimensionality, constitutes the intrinsic macroscopic-thermodynamic limiting phase of this discrete computational model.

Methodological Supplementary Remark (added in v1.1)

Multidimensional Scaling (MDS) serves only as an offline post-processing numerical prototype. MDS reads static snapshots of the binary network and embeds pairwise topological relations into three-dimensional Euclidean space. Spatial distances within this embedded manifold are emergent representations of node relational-profile similarity; **Euclidean geometric distance does not pre-exist within the underlying binary-network ontology**. This static-snapshot embedding is conceptually distinct from the event-driven, incremental-instantiation mechanism characterising the full SRE-Dynamics physical-rendering-layer. Embedding artefacts appear in strongly-dissipative outer regimes beyond the coherence length ξ , where the validity of low-dimensional geometric descriptions degrades.

Appendix A Conceptual Distinctions and Applicable Boundaries for MDS Numerical Embedding (added in v1.1)

A.1 Conceptual Distinction between MDS Simulation Prototype and the SRE Physical-Rendering-Layer

Within the accompanying simulation script `sim_P.py`, Multidimensional Scaling (MDS) operates as **snapshot-oriented offline post-processing**: after full network evolution to a specified step count, the complete final-state matrix is taken, and pairwise topological relations for all nodes are projected in a single pass onto three-dimensional Euclidean coordinates.

This numerical workflow possesses two key features:

1. It is not incremental-difference rendering: only complete snapshots of all node-topology relations are

input; only differential changes per evolutionary step are not rendered.

2. It performs homomorphic dimensional reduction rather than isomorphic copying: the underlying object is high-dimensional discrete binary topology possessing no Euclidean coordinates. MDS executes many-to-one coarse-graining and information compression. Numerous microscopic degrees-of-freedom including underlying spin signs, activate-dormancy stochastic gates, and local frustration are not fully preserved within three-dimensional output.

By contrast, the **physical-rendering-layer** defined in the SRE v1.6 ontological framework is an event-driven incremental-instantiation system:

The underlying causal network undergoes continuous asynchronous-link fluctuations. The vast majority of transient perturbations fail instantiation-cost thresholds and remain in the pending-for-instantiation state without projection onto spacetime. **Only causal relations completing one full round of reciprocal measurement together with dissipation-compensation settlement update distance- and time-bookkeeping records within the rendering layer.**

MDS simulation: static characterisation of an already-evolved system, answering what three-dimensional manifold would arise via homomorphic projection given statistically stable topology.

SRE physical-rendering-layer: real-time incremental instantiation triggered by reciprocal-measurement events; the ontological mechanism corresponding to real-world physical cosmology. The two must not be treated as directly equivalent.

A.2 Intuitive Rules and Failure Conditions for MDS Manifold Distances

Geometric distances output by MDS embedding follow this statistical tendency:

If two nodes exhibit highly similar patterns of causal association (relational-profiles) with all remaining network nodes, their Euclidean distances within the three-dimensional embedded manifold tend to be small.

This property corresponds to SRE ontological intuition: macroscopic spatial distance characterises the degree of coherence degradation between events. More similar relational-profiles imply lower mutual information dissipation, smaller topological-compensation cost, and correspondingly smaller macroscopic metric distance.

This rule-of-thumb has strict applicability boundaries:

- **Microscopic core region** ($x \leq \xi$, **high order-parameter** $\Phi(N)$): high network topological coherence yields good MDS dimensional-reduction fidelity; the relational-profile versus manifold-distance correspondence holds.
- **Mesoscopic / macroscopic strong-dissipation regimes** ($x > \xi$): abundant stochastic link dormancy creates highly disordered topology; no low-dimensional geometry can perfectly represent all high-dimensional relations. MDS produces minimal-loss approximate fits and embedding artefacts may occur: nodes with substantially different underlying relational-profiles may lie accidentally close geometrically, while minor topological differences may be artificially amplified during dimensional compression. Under these conditions low-dimensional geometric descriptions degrade, and the relational-profile / manifold-distance correspondence becomes unreliable.

A.3 Correspondence with the Full SRE System

Phenomena within the binary-network simulation can be mapped in parallel to SRE ontological concepts, though direct identification is not permitted:

1. Link dormancy $\chi_{ij} = 0 \Leftrightarrow$ irreversible information dissipation (microscopic origin of the topological-dissipation tensor $\hat{\mathcal{D}}_{ij}$ in cosmological papers).
2. MDS dimensional-reduction embedding $\Leftrightarrow$ numerical demonstration prototype for multi-scale homomorphic mapping.
3. Core / meso-scale / outer-layer tripartite division $\Leftrightarrow$ multi-scale rigid coherence-truncation boundary filtering mechanism.
4. Finite-size-scaling phase transition $\Leftrightarrow$ underlying discrete prototype for the BBP spectral phase transition ($z_{\text{crit}} = 4.1605$, 2D-holographic $\leftrightarrow$ 4D-spacetime switching).

Note: the binary network itself contains no built-in spacetime, gravitational constant, or speed of light. All such physical quantities are macroscopic statistical effects emerging inside the rendering-layer after homomorphic-mapping filtering.

In summary, this binary network constitutes the discrete computational substrate for SRE's self-emergent cosmic picture. Full cosmological physical predictions require ontological abstraction plus spectral-matrix mathematical upgrades; refer to the accompanying SRE v6.0 paper.

A.4 Version-Change Notes

v1.1 update record:

1. Added brief MDS methodological warning at the end of Chapter 5 Theoretical Conclusion.
2. Added Appendix A giving full conceptual distinction between the MDS numerical prototype and the SRE physical-rendering-layer, together with applicable-boundary discussion.
3. The v1.0 manuscript is fully preserved in the Zenodo archive for historical traceability and is not overwritten.